



Medical Microbiology

Lecture one

Morphology & ultra – structures of microbiology

Third stage- College of Dentistry

Dr. Dheyaa Naji Hamza

Dheyaa.naji@alkafeel.edu.iq

Q/Why is microbiology important in dentistry?



Microbiology is important in dentistry

- Because it helps identify the microorganisms responsible for oral diseases like **dental caries** and **periodontal infections**.
- Understanding these microbes allows dentists to diagnose, prevent, and treat oral infections effectively.



Microbiology:

Microbiology is the branch of science that focuses on microorganisms, which are not visible to the naked eye.

Microorganisms include: bacteria, viruses, fungi, and parasite, (protozoa and Helminthes).



Eukaryotic and Prokaryotic cells:

Based on the organization of their cellular structures, all living cells can be divided into two type of cells:

1. Cells that have membrane-bound organelles Called **Eukaryotic Cells** **that include:** Animals, plants, fungi, protozoa, and algae.
2. Cells that do not have membrane-bound organelles called **prokaryotic cells** **that include:** Unicellular organisms such as bacteria and blue green algae.

Eukaryotic cell:

Eukaryotic cells tend to be 10 to 100 times the size of prokaryotic cells and includes the following organelles:

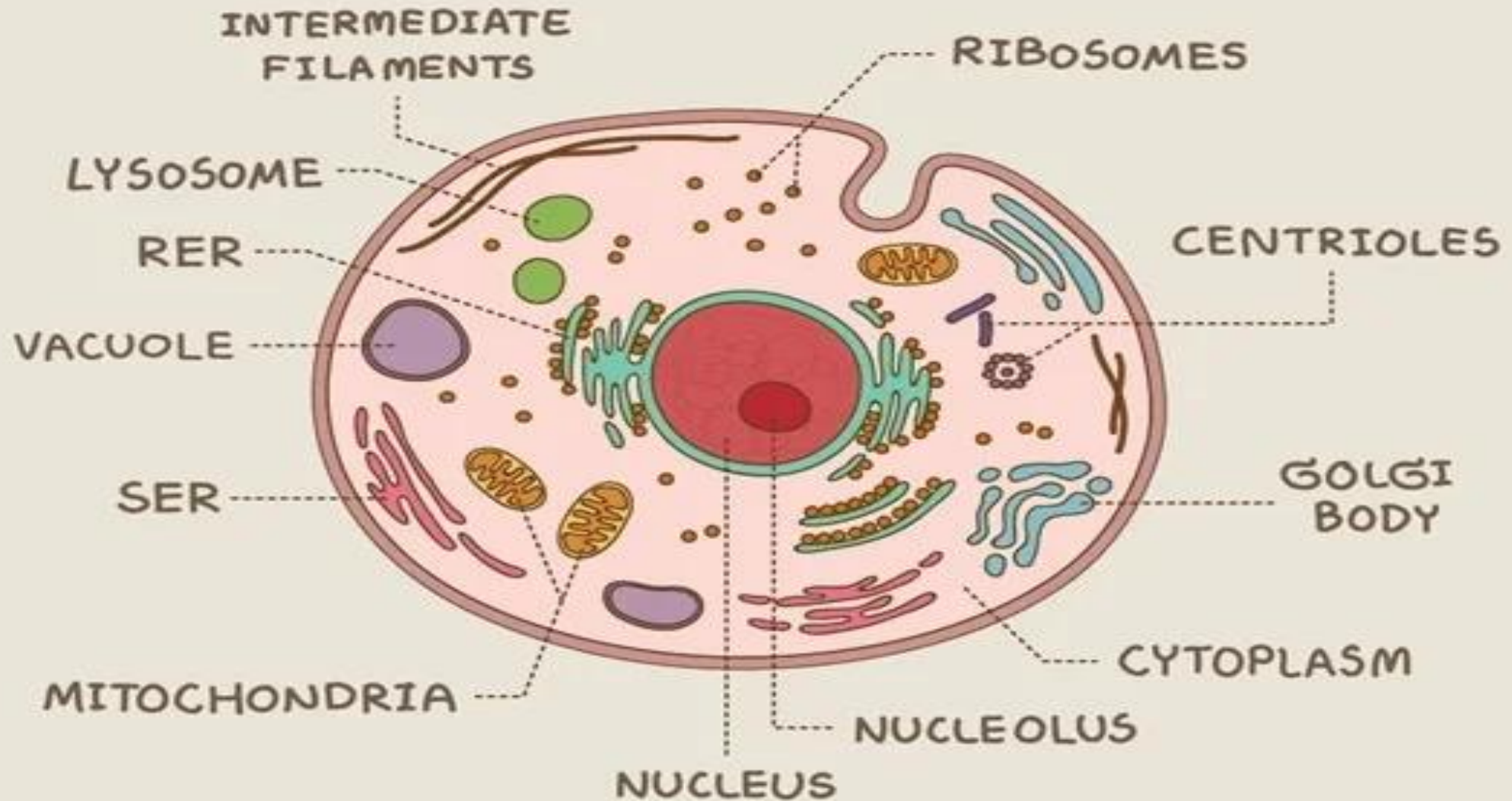
1-Cytoplasm of eukaryotic cell is a jelly-like material that contains dissolved nutrients, salts and structures called **organelles**. It is where many of the chemical reactions happen like glycolysis and protein synthesis.

2-Chromosomes are located within the nucleus, a membrane-bound organelle that contains the instructions for protein synthesis and regulates the cell's activities.

3-Cell membrane is also called the **plasma membrane**, is found in all cells and separates the interior of the cell from the outside environment and permeable to some substances but not to others. It therefore controls the movement of substances in and out of the cell.

4-Its also contains membrane-bound organelles such as **Golgi bodies, Mitochondria & chloroplast**, there are metabolic function and energy conversion .

5-Eukaryotic cell are two types of cell division : **sexual & Asexual**.

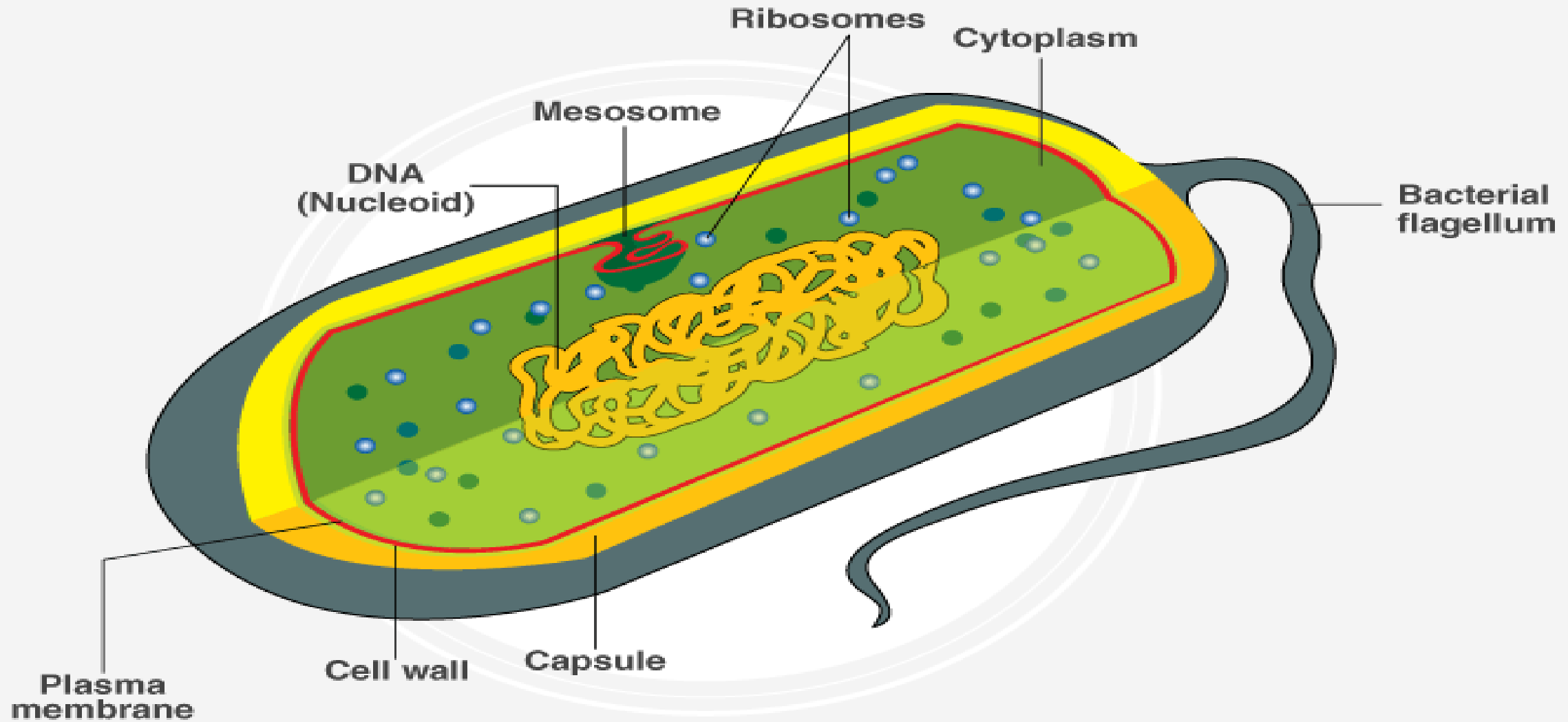


Eukaryotic cell

Prokaryotic cell:

- Prokaryotes are evolutionarily **ancient**. These cells are simpler than eukaryotes and **don't contain a nucleus** and These cells had chromosomal DNA is a circular loop & a plasmid .
- They have few internal structures in the cytoplasm on this group.
- Its also do not have membrane-bound organelles .
- The cell division occur by a process called **binary fission**.

PROKARYOTIC CELLS



Differences between prokaryotic & eukaryotic cells

Prokaryotic

- no membrane bound organelles
- no true nucleus
- unicellular
- 0.1-5 micrometers
- has cell wall
- asexual reproduction

- Ribosomes
- Cell membrane
- Has DNA
- Cytoplasm

Eukaryotic

- contains membrane bound organelles
- contains true nucleus
- uni-, multicellular
- 10-100 micrometers
- asexual and sexual reproduction

Bacteria:

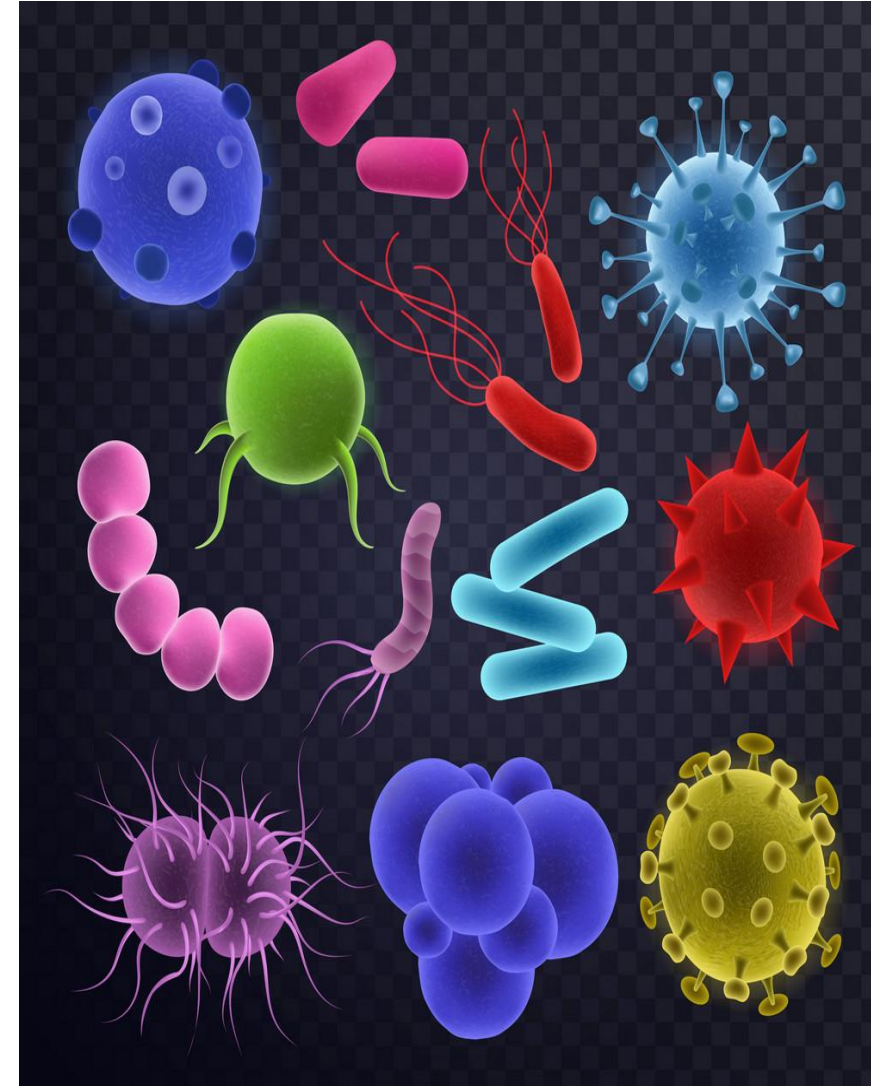
It is prokaryotic microorganism can be seen easily by using microscope . it is unicellular free living organisms ,they are capable of performing all essential processes like metabolism and reproduction.

Size of Bacteria:

Unit of measurement in bacteriology is the micron / micrometer (μm)

Bacteria of medical importance

- (0.2 – 1.5) μm in diameter
- (3 – 5) μm in length



Shape of Bacteria:

Cocci – spherical/ oval shaped (major groups)

Bacilli – rod shaped

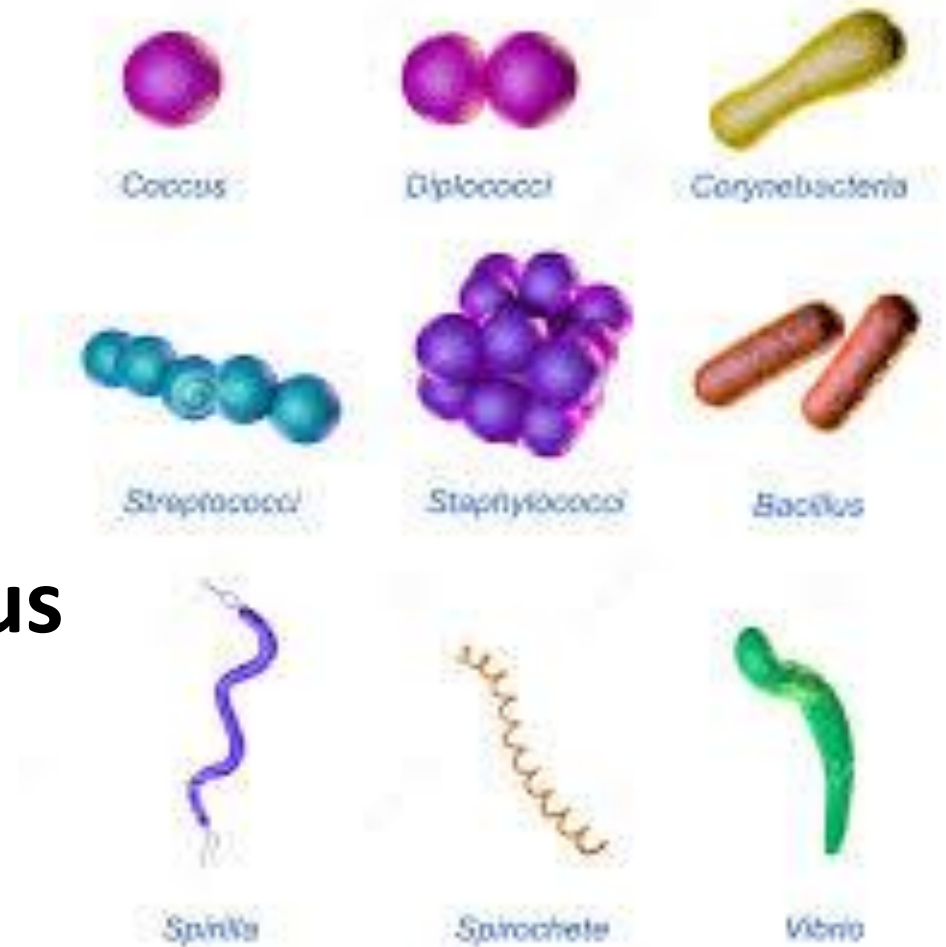
Vibrio – comma shaped

Spirilla – rigid spiral forms

Spirochetes – flexible spiral forms

Actinomyces – branching filamentous

Pleomorphic - variable in shape



Bacterial cell structure:

organized into 3 categories :

- **Internal Structures:** Cytoplasm, bacterial chromosome, plasmid, ribosomes, and storage granules.
- **Cell envelope:** cell membrane, peptidoglycan cell wall, or an outer lipid membrane (only found in Gram-negative cells).
- **External structures :** flagella, fimbriae, sex pilus , Capsule and spore.

Bacterial Internal Structures:

A-Cytoplasm:

Cytoplasm is Colloidal system of variety of organic & inorganic solutes in viscous watery solution . It helps in cellular growth, metabolism and replication.

Cytoplasmic Components :-

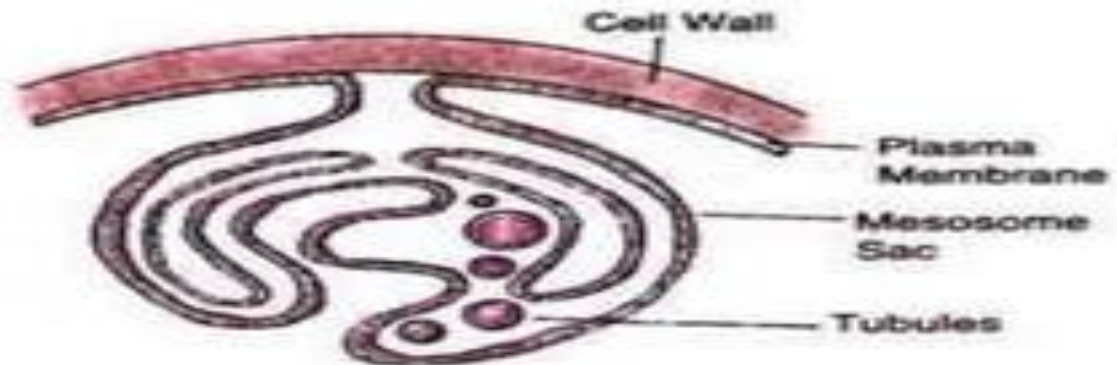
1-Ribosomes:-

Ribosomes are the sites of protein synthesis. Bacterial ribosomes differ from those of eukaryotic cells in both size and chemical composition. They are organized in units of 70S in prokaryotic , compared with eukaryotic ribosomes of 80S.

2. Mesosomes:-

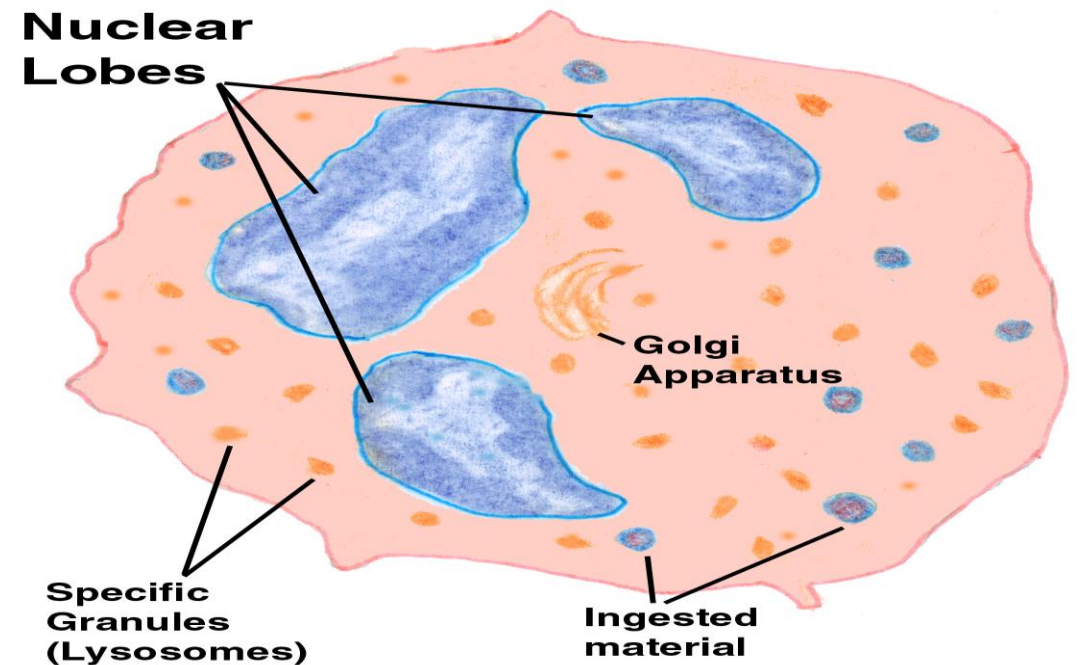
- Multi-laminated structures formed as invaginations of plasma membrane.
- Principal sites of respiratory enzymes.
- Coordinate nuclear and cytoplasmic division during binary fission.
- Responsible for compartmenting DNA during sporulation.

Mesosome diagram



3-Cytoplasmic granules :

Many bacteria that may occupy a substantial part of the cytoplasm to form granules . Some bacteria accumulate in organic phosphate substance which is called volutin granules which is metachromatic granule e. g (*Corynebacterium diphtheriae*). It is a reserve of energy & phosphate for cell metabolism.



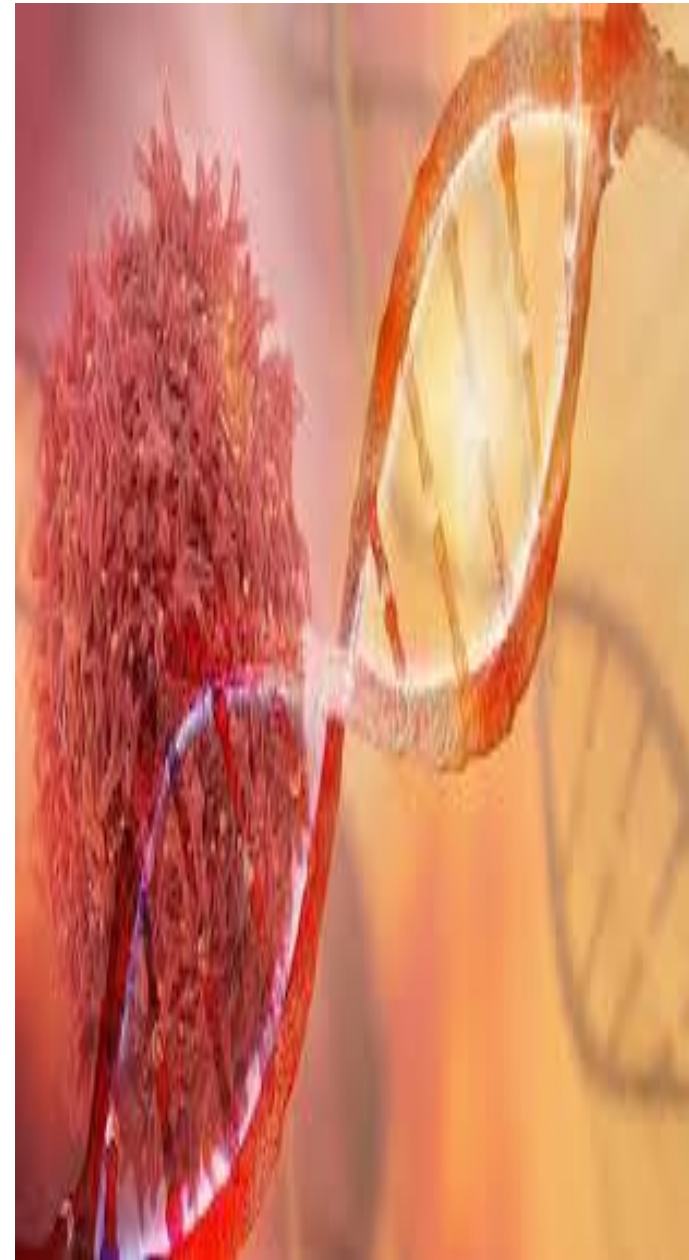
B- Chromosome (nucleic acid):

- bacteria do not have a nucleus. Instead, the bacterial chromosome is found floating within a dense region of the cytoplasm called the **nucleoid region**.
- Bacteria only have chromosomes , which are tightly-coiled, circular pieces of DNA that contain all of the genetic information required for growth & survival of bacteria.
- Bacteria contains about 1000 genes are located in **circular chromosome**.

Structures of Genetic material of bacteria (bacterial genome):

1- DNA(Deoxy ribose Nucleic acid):

- In the 1950's, Watson & Crick built the first model of DNA.
- DNA composed of two strands are stabilized by hydrogen bound to form double helix.
- DNA molecule has simple unite called nucleotides ,which contain four nitrogenous base:
- **purine** (adenine(**A**) & guanine(**G**)) and
- **Pyrimidine** (Thymine (**T**)&Cytosine(**C**)) are attached to sugar called **deoxyribose** .
- There is a back bone of alternating deoxyribose and phosphate .



2- RNA(ribose Nucleic acid):

Structurally similar to DNA except :

1-Ribose sugar instead of deoxyribose.

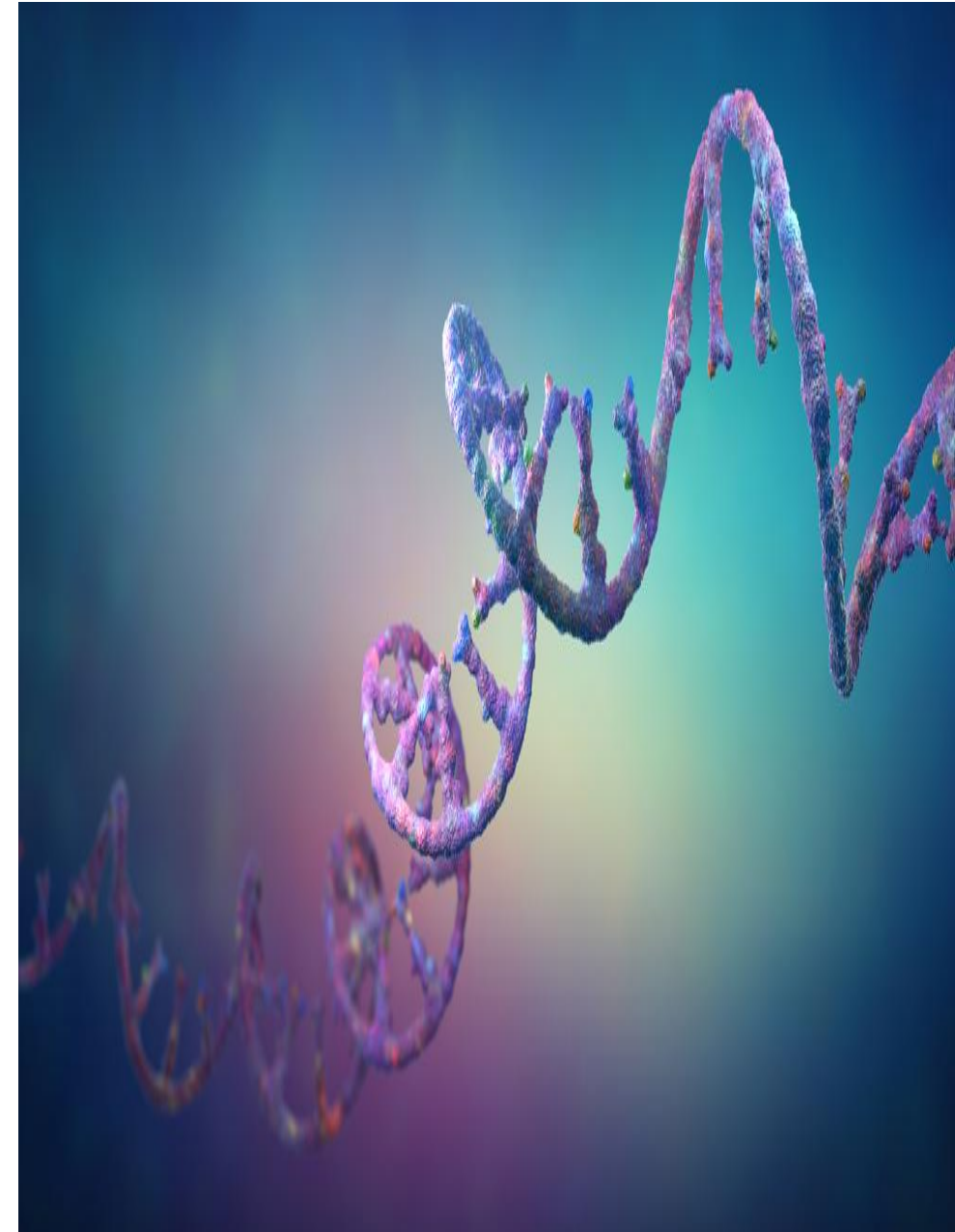
2-Uracil replaces thymine.

3- Three types of RNA

Messenger RNA (mRNA)

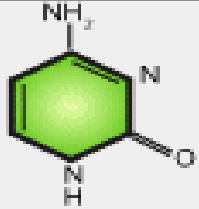
Ribosomal RNA (rRNA)

Transfer RNA (tRNA)

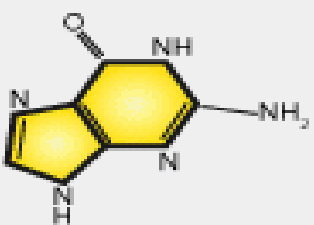


DIFFERENCE BETWEEN DNA AND RNA

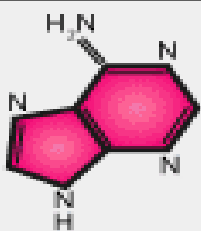
CYTOSINE C



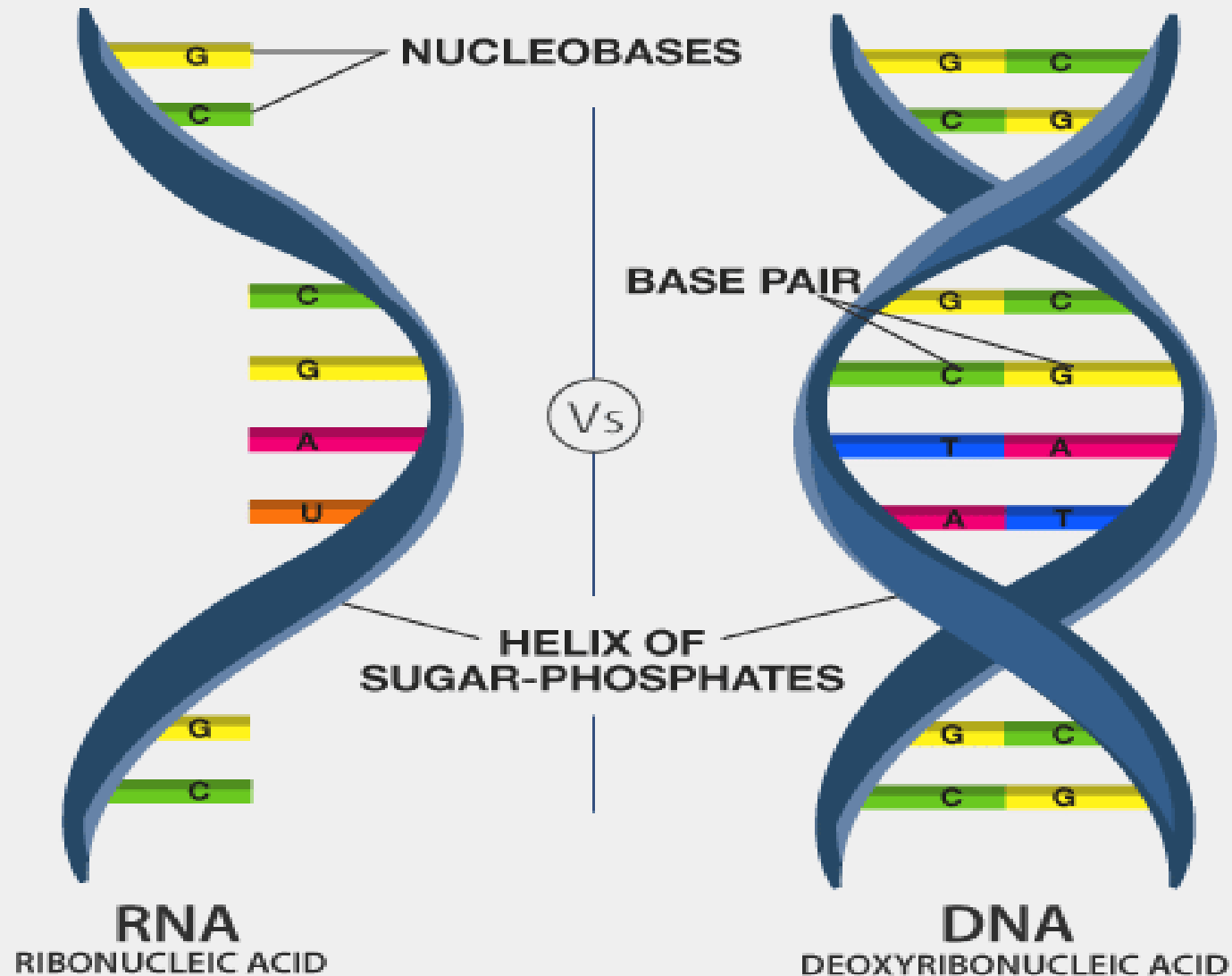
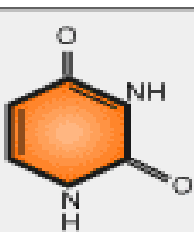
GUANINE G



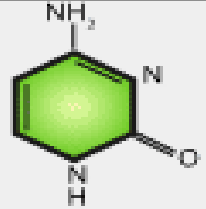
ADENINE A



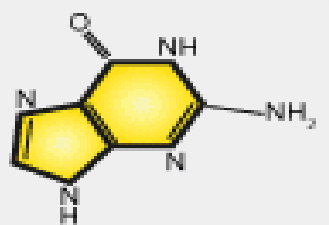
URACIL U



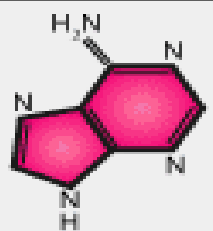
CYTOSINE C



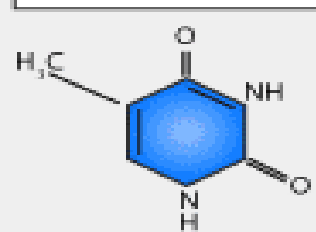
GUANINE G



ADENINE A



THYMINE T



3-Plasmid :

1-The plasmid usually extra-nuclear genetic elements consisting of circular DNA present the cytoplasm of the bacteria.

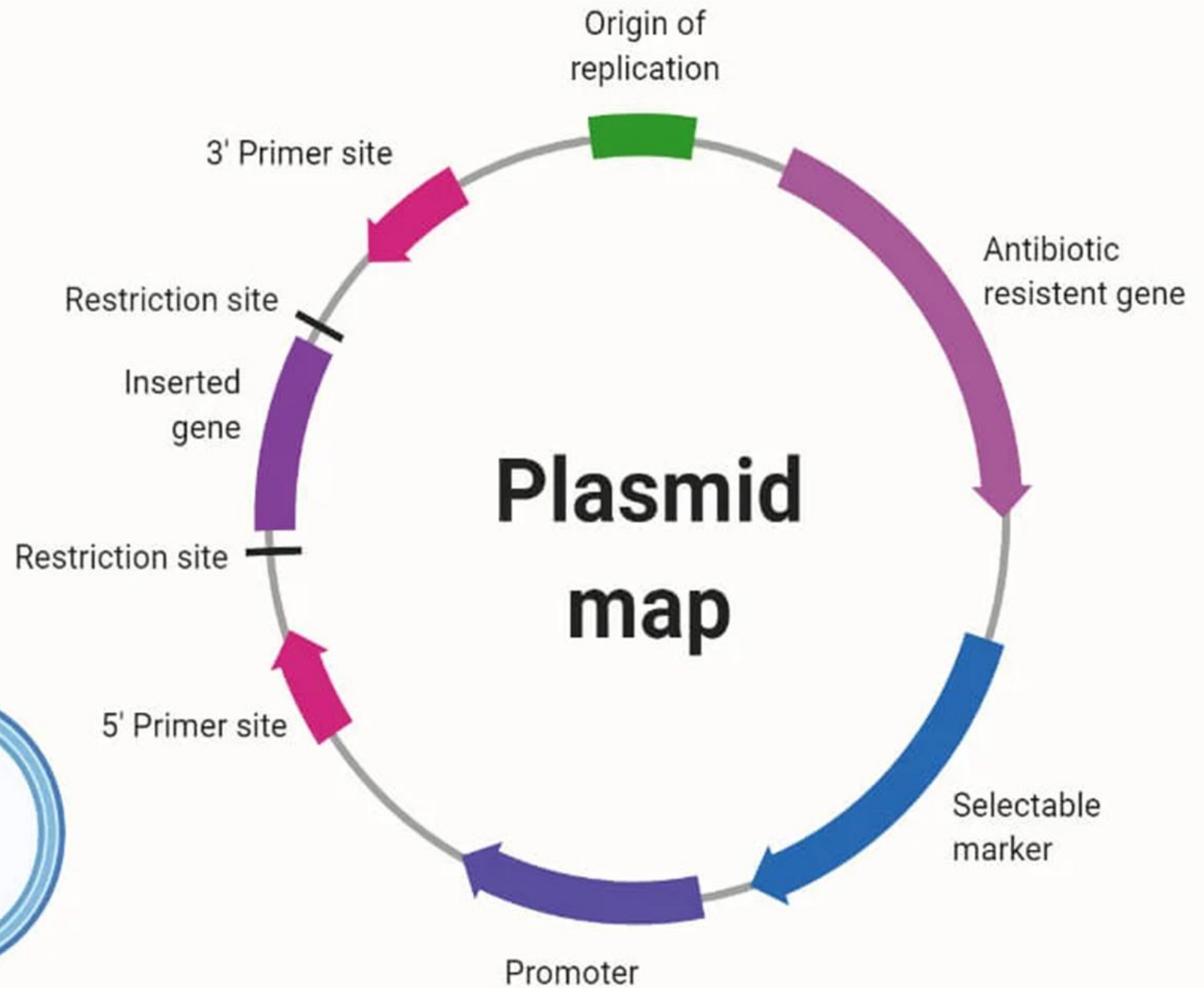
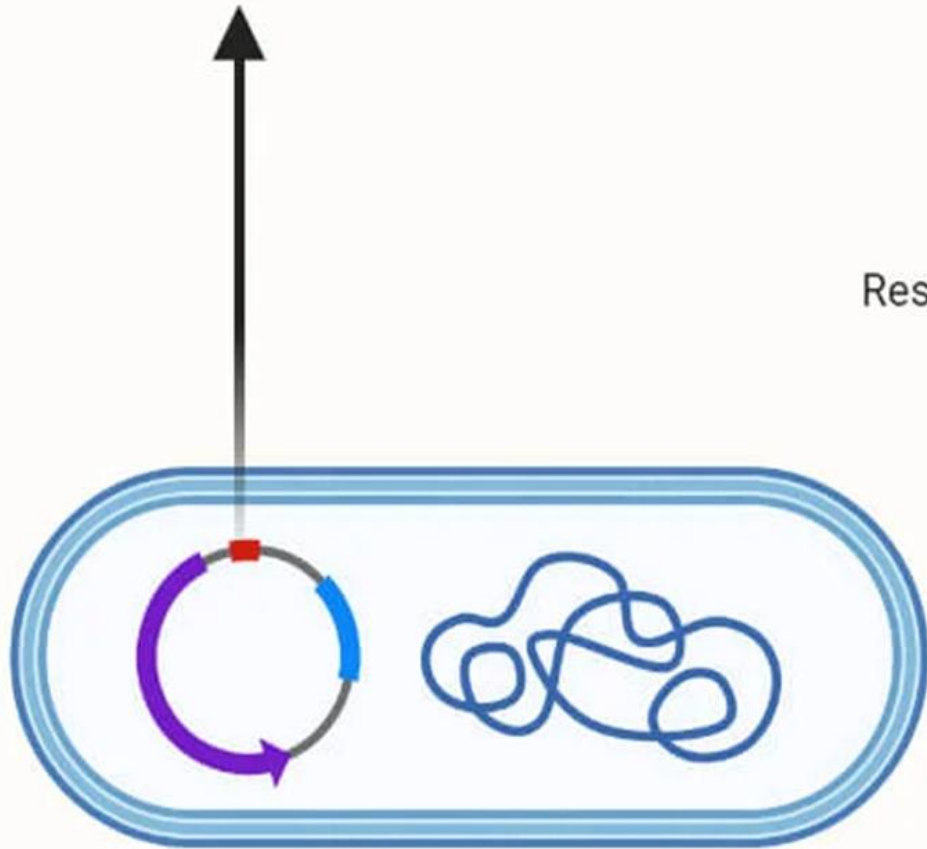
2- Transmitted to daughter cells during binary fission.

3- May be transferred from one bacterium to another by **conjugation**.

4-Its also can perform some additional activities to the bacteria, plasmid code gene for enzymes , toxin or resistance factors to antibiotics and transmit these pathogenic factors from bacteria to another .



Plasmid



Cell envelope :

1-Cell membrane : the semipermeable membrane surrounding the cytoplasm of a cell.

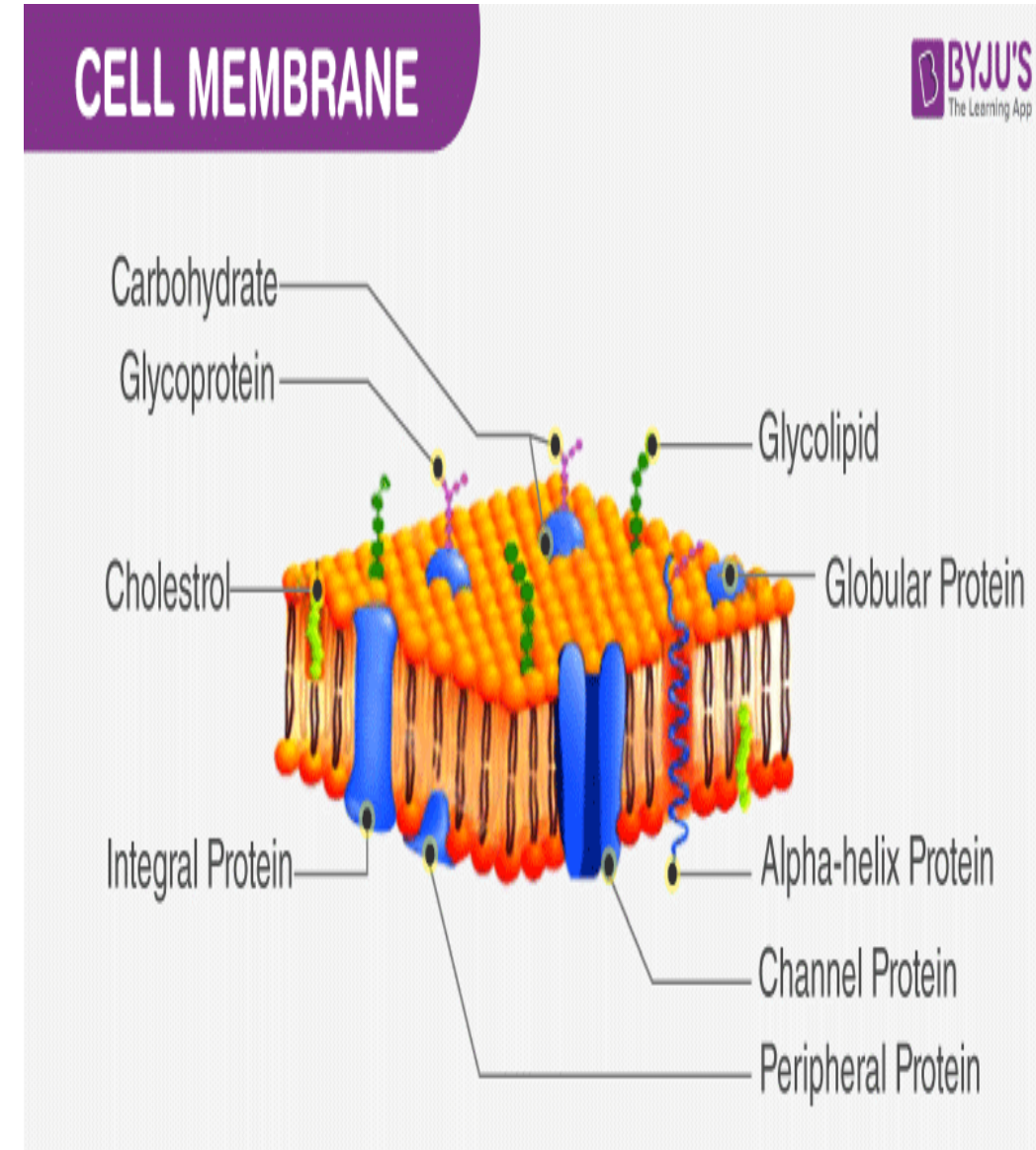
Function of cell membrane:

1-Protecting the integrity of the interior cell.

2-Providing support and maintaining the shape of the cell.

3-Helps in regulating cell growth through the balance of endocytosis and exocytosis.

4-The cell membrane also plays an important role in cell signaling and communication.

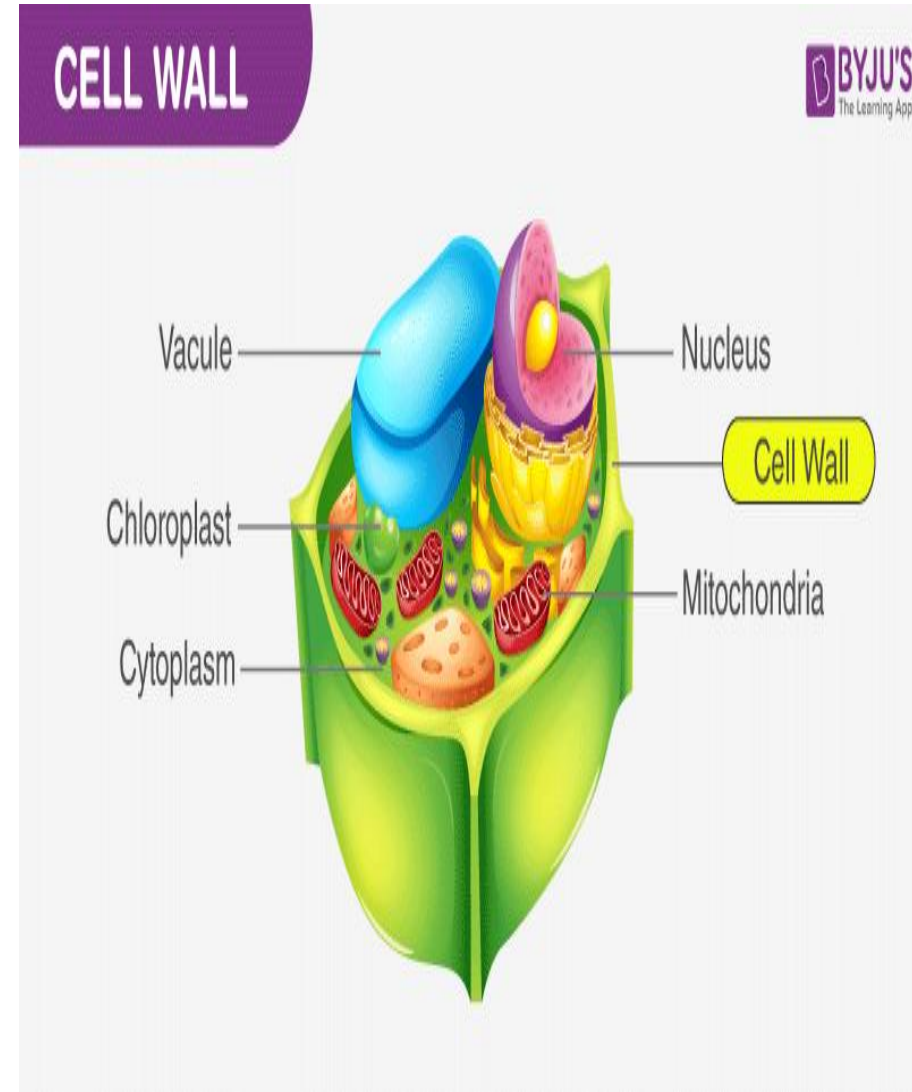


2- Cell wall :

a rigid layer of polysaccharides lying outside the plasma membrane of the cells of plants, fungi, and bacteria.

Function of cell wall:

- 1-maintains cell shape .- Acts as a barrier, protects cell contents from external environment .
- 2- maintains cell integrity/osmotic pressure in a hypotonic environment.
- 3 - Determines reactivity to Gram stain.
- 4 - Attachment site for flagella.
- 5 -Contributes to sensitivity to certain antimicrobial agents and the immune system (antibodies, phagocytes).



External structures:

1-Capsule or slime layer : The capsule is a gelatinous layer covering the entire bacterium.

It is composed of polysaccharide except in the anthrax bacillus which composed of

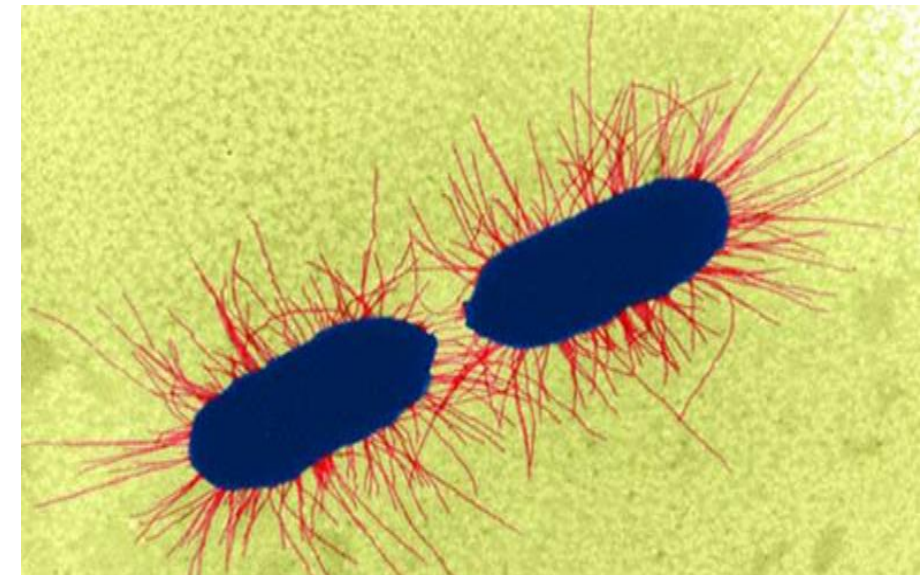
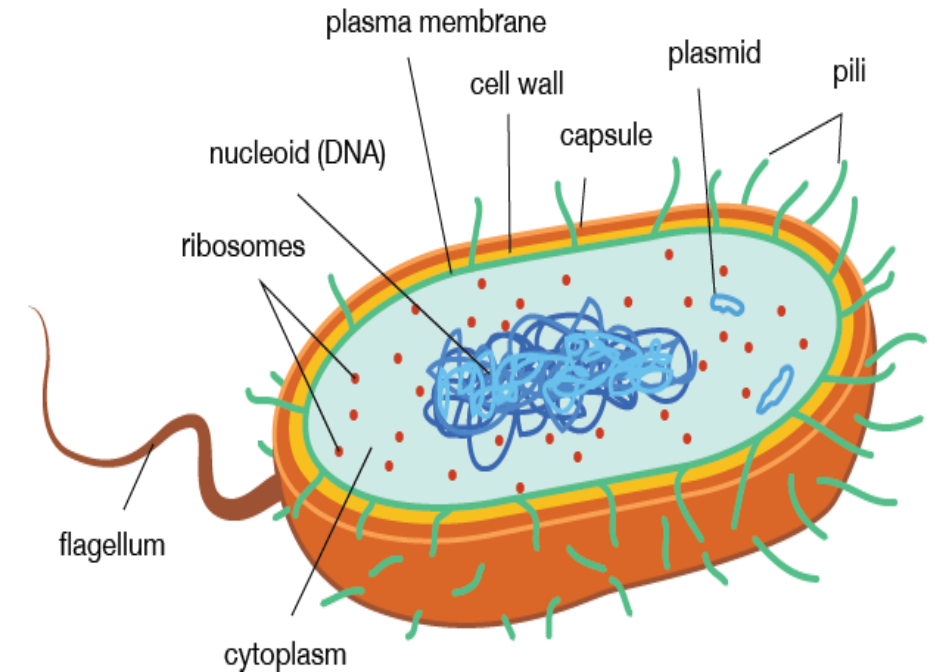
D-glutamic acid.

It is a determinant of **virulence** of many bacteria since it resist **phagocytosis** and **attachment** of host tissues.

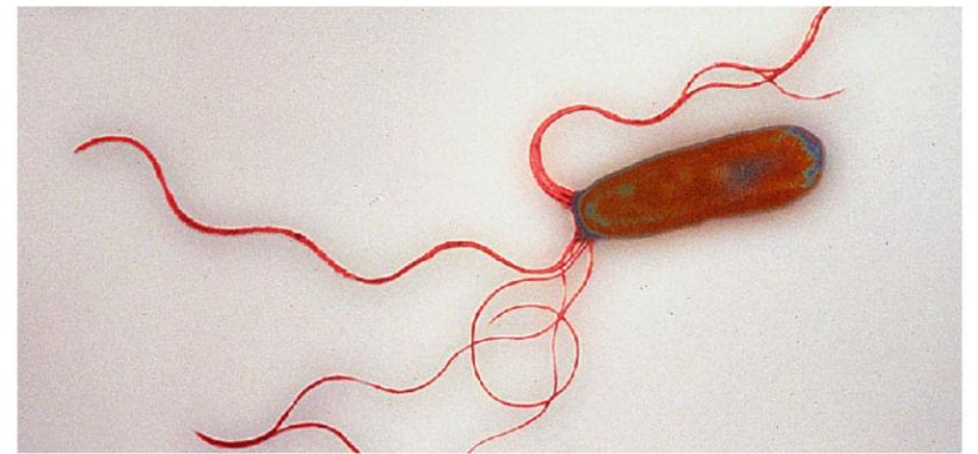
2- Pilus (fimbria) : Hair-like short filamentous **not for motility**, are numerous extend from cell surface.

Consist of protein mainly found in Gram-negative.

It is necessary for bacterial **conjugation** resulting in the transfer of DNA from one cell to another.



3- Flagellum : a slender threadlike structure, especially a microscopic whip like appendage that enables many protozoa, bacteria, etc., to swim.



Arrangement basis for classification:

Monotrichous; 1 flagella

Lophotrichous; tuft at one end

Amphitrichous; both ends

Peritrichous; all around bacteria



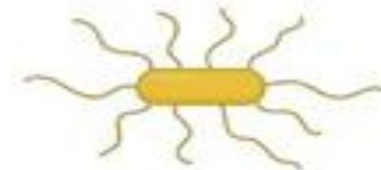
Monotrichous



Lophotrichous



Amphitrichous



Peritrichous

Fig:Types of flagella

Created in BioRender.com bio

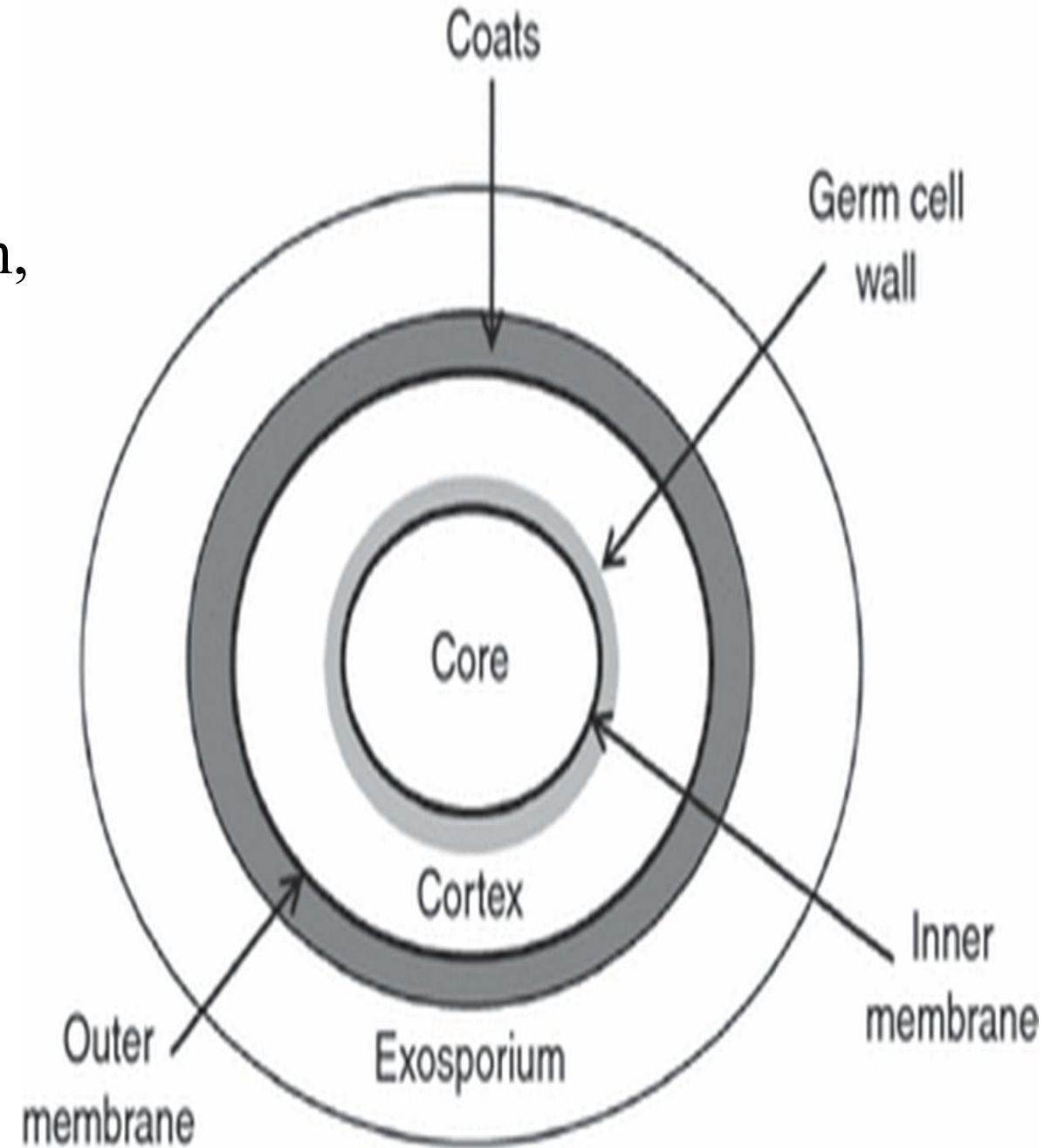
4-Spores:

1-Permit the organism to survive during conditions of desiccation, nutrient depletion, and waste buildup.

2-Have low-water content with incorporation of Ca- dipicolinic acid.

3-Highly resistant to disinfectants, antibiotics, radiations, boiling....etc.

Requires heating to 120°C for 15-20 min.





**Any
Question**

Thank you